The infinite-breadth weighted-semigroup question was answered affirmatively

Literature-status packet for arXiv:1901.00082

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Status. Literature already answered: full affirmative answer.

Source question

Yemon Choi, Mahya Ghandehari, and Hung Le Pham ask in Section 4 of *Stability of characters and filters for weighted semilattices* (arXiv:1901.00082; source PDF page 13):

> Let S be a semilattice with infinite breadth. Does there exist a log-weight on S for which propagation fails at some level?

Their Theorem 4.8 answers this only when S is isomorphic to a subsemilattice of the finite-subset semilattice $(\mathrm{FIN}(\Omega), \cup)$. By their AMNM–propagation equivalence, the full question is equivalent to asking whether every infinite-breadth semilattice admits a submultiplicative weight ω for which $\ell^1(S, \omega)$ is not AMNM.

Exact later answer

The same three authors return to this exact question in *Constructing non-AMNM weighted convolution algebras for every semilattice of infinite breadth* (arXiv:2305.18272). In its introduction they quote the general question, identify their earlier Theorem 4.8 as a partial answer, and explicitly say that the new article extends that partial answer to a full answer. Their Theorem B on supporting PDF page 3 states: if S has infinite breadth, then one can construct a submultiplicative weight ω such that $\ell^1(S, \omega)$ is not AMNM.

Thus the answer to the source question is affirmative for every semilattice of infinite breadth, with no additional representability hypothesis on S.

Scope and provenance

This packet records a literature result and does not claim a new proof. The match is exact rather than inferred: the supporting authors are the source authors, cite the source theorem, reproduce the unrestricted question, and call their theorem a full answer. Hence no part of the stated source question remains open.

Search evidence

The run's lightweight indexes were checked for the source arXiv identifier and the terms "infinite breadth", "propagation", and "non-AMNM". An exact-title and keyword search then located arXiv:2305.18272. The source question on PDF page 13 and the supporting paper's identification and Theorem B on PDF page 3 were checked in the locally compiled source PDFs.

References

1. Y. Choi, M. Ghandehari, and H. L. Pham, *Stability of characters and filters for weighted semilattices*, Semigroup Forum 102 (2021), 86–103; arXiv:1901.00082.

2. Y. Choi, M. Ghandehari, and H. L. Pham, *Constructing non-AMNM weighted convolution algebras for every semilattice of infinite breadth*, arXiv:2305.18272.